

CSA0609-DESIGN AND ANALYSIS OF ALGORITHMS

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Exp-01: First Palindromic String in an Array and Find its Complexity

Aim:

To find the first palindromic string in an array and analyze its time and space complexity.

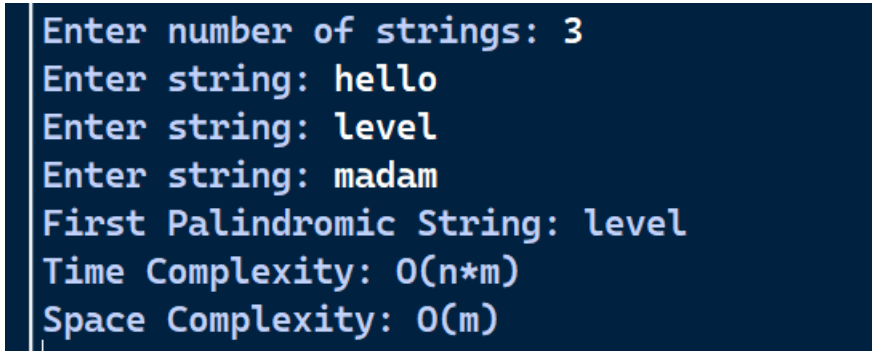
Algorithm:

1. Read the number of strings.
2. Read the strings into an array.
3. Check each string one by one.
4. Reverse the string and compare it with the original.
5. If both are equal, display the first palindromic string.
6. Display the time and space complexity.

Source Code:

```
n = int(input("Enter number of strings: "))
arr = []
for i in range(n):
    arr.append(input("Enter string: "))
found = False
for word in arr:
    if word == word[::-1]:
        print("First Palindromic String:", word)
        found = True
        break
if not found:
```

```
print("No Palindromic String Found")  
print("Time Complexity: O(n*m)")  
print("Space Complexity: O(m)")
```

Output:A screenshot of a terminal window with a dark blue background and light blue text. The text shows the user inputting three strings: 'hello', 'level', and 'madam'. The program identifies 'level' as the first palindromic string and displays its time and space complexities.

```
Enter number of strings: 3  
Enter string: hello  
Enter string: level  
Enter string: madam  
First Palindromic String: level  
Time Complexity: O(n*m)  
Space Complexity: O(m)
```

Result:

The first palindromic string was successfully found and its complexity was analyzed.